

Enabling Connectivity

Jonathan Morris
General Manager
Fabric IP Business Unit

Enterprise Solutions



HOME
SOLUTIONS



ENTERPRISE
SOLUTIONS



MOBILE
SOLUTIONS



EMBEDDED
SOLUTIONS

EMERGING APPLICATIONS

Enabling Connectivity



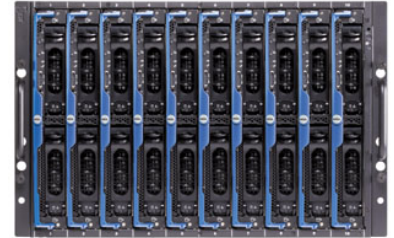
Wireless
Networking



Voice over IP



ENTERPRISE SOLUTIONS



Blade Servers



Portable Content



Home Networking



Small – Medium
Business (SMB)



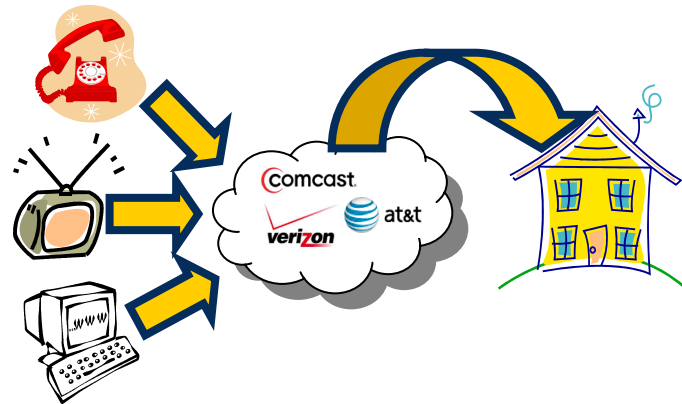
Printers



Mass Storage

New Challenge in Consumer Networks

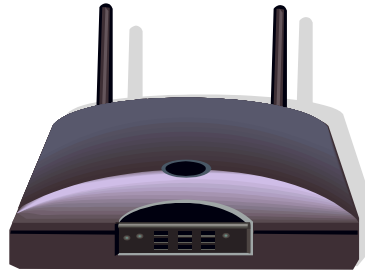
- High-performance connectivity is no longer a feature - it's a requirement
 - “Triple Play” of Voice, Video and Data through a single network pipe drives bandwidth needs up
 - Consumer thirst for bandwidth keeps growing
 - Fueled by Everything-over-IP phenomenon
 - Quality of Service (QoS) becomes major consideration



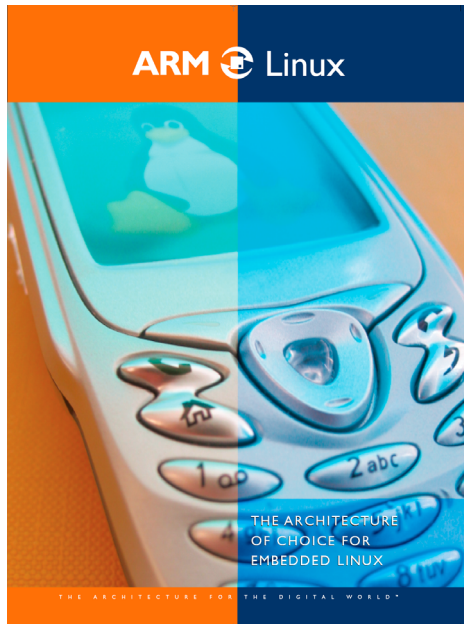
- A familiar problem in Carrier Class networks,
 - Need to find a scalable mass-market solution

Critical Success Factors

- Cost Effective
 - Power efficiency needed for package cost and fan elimination
- Standard S/W Infrastructure
 - Complete Networking Stack, including NAT, IPSec, Firewall,
- Compiler and Debug Tools
 - Development tools strongly integrated with Operating System
- Scalable Performance
 - Design headroom for evolving standards
- End-to-end QoS
 - Intelligent fabric to ensure traffic prioritisation
- Right-first-time Design
 - Extensive pre-silicon design validation of “random” packet flows and conflicts through system modeling



ARM: First Choice for Embedded Linux



Board Support

ARM Linux Distribution

GNU Tools

RealView® Tools

Linux Applications

debian

CODESOURCERY

handhelds.org

montavista™

WIND RIVER

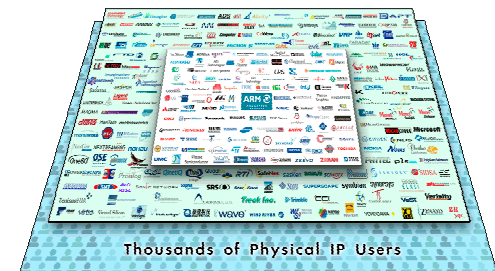
OSDL



CE Linux Forum

REQUIREMENT	ARM LINUX	OPEN SOURCE LINUX	COMMERCIAL LINUX
Access to early ports compatible with latest ARM architectures	   	 	 
Range of ports for ARM cores and boards	  More core support	   	  More board support
Standard model for Linux services — support and consulting	 		   
Training			   
Tools and Development Environments	   	 	   

LIPS
Linux Phone Standards Forum



Proven Linux Solutions

■ Large installed base of ARM Powered Linux products

- Proven stability and reliability
- Wide use in networking and infrastructure
- Unified APIs: reuse existing software

■ ARM commitment to open source

- Maintain kernel in sync with latest developments
- Broadest choice of Linux solutions
- Pre-built Linux Board Support Packages and GNU GCC tools

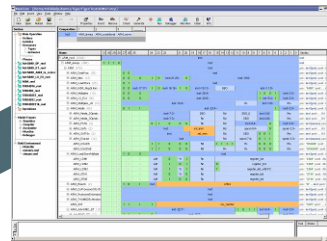
■ Optimised software solutions for Linux

- Power, Security, SMP, Java

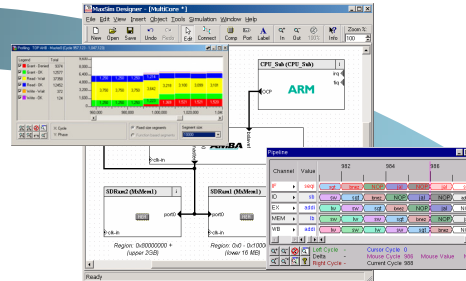
Linux Board Support Package Availability

Linux OS Source	Core Support	Board Support
ARM EMBEDDED LINUX www.arm.com	ARM Linux is supported on ARM cores which have an MMU, including: ARM720T™ ARM920T™ ARM922T™ ARM926EJ-S™ ARM10 Family ARM1136JF-S™ ARM1136JF-S™ ARM1176JF-S™ ARM11 MPCore™ ARM Cortex Family	REALVIEW® VERSATILE PLATFORMS Platform based on a highly integrated development chip, supporting systems based on ARM7TDMI, ARM926EJ-S, ARM1136JF-S and ARM11 MPCore. www.amba.com/products/DevTools/VersatileFamily.html REALVIEW INTEGRATOR® PLATFORMS Prototyping platform supporting systems based on ARM922T, ARM926EJ-S, ARM946E-S, ARM1026EJ-S and ARM1136JF-S. www.amba.com/products/DevTools/IntegratorFamily.html REALVIEW SOFT MARCROCELL MODELS RealView® Soft Macrocell Models provide an ideal platform for evaluating, benchmarking and prototyping systems based on ARM processors. www.amba.com/products/DevTools/SMMs.html
LINEO SOLUTIONS INC. www.lineo.co.jp	ARM920T™ XScale™	Yokogawa A-Plat (DragonBall i.MX1) Intel® DBPXA250
MONTAVISTA SOFTWARE www.mvista.com	ARM720T ARM920T ARM926EJ-S ARM1136JF-S Intel® XScale® Intel StrongARM	Altera Excalibur EPXA1 RealView Versatile Platform Baseboard for ARM926EJ-S Freescale Semiconductor i.MX development boards Intel PXA and IXP boards Philips PXN boards Samsung SMDK boards TI OMAP™
OPEN SOURCE ARM LINUX www.arm.linux.org.uk	This web site currently lists Linux ports to over 400 ARM-based platforms and host ARM Linux open source newsgroups.	
TIMESYS® www.timesys.com	ARM920T ARM926EJ-S	RealView Integrator Freescale i.MX21
UCLINUX www.uclinux.org	ARM7TDMI ARM946E-S ARM1156T2(F)-S	ATMEL ARM7TDMI
WIND RIVER www.windriver.com	ARM926EJ-S ARM1136JF-S Intel XScale	RealView Versatile Application Baseboard for ARM926EJ-S

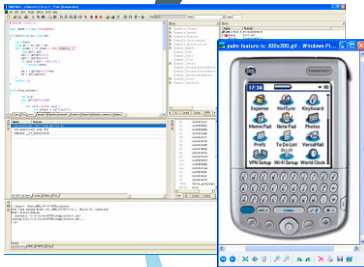
Comprehensive Tools



IP Creation

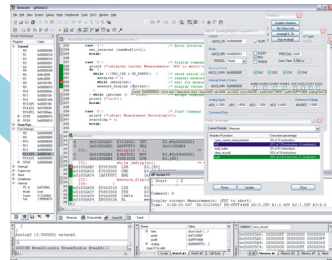


Architectural Exploration

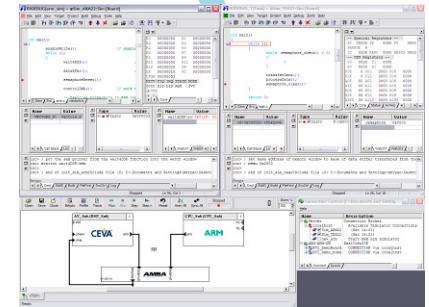


Applications Development

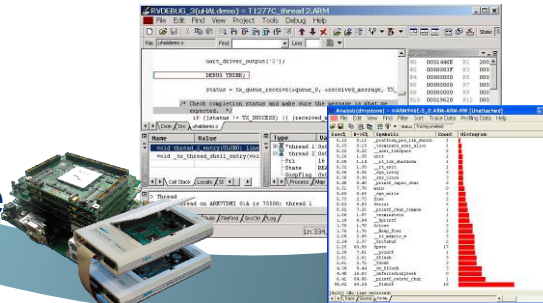
RealView®
Tools by ARM®



Microcontroller Software Development

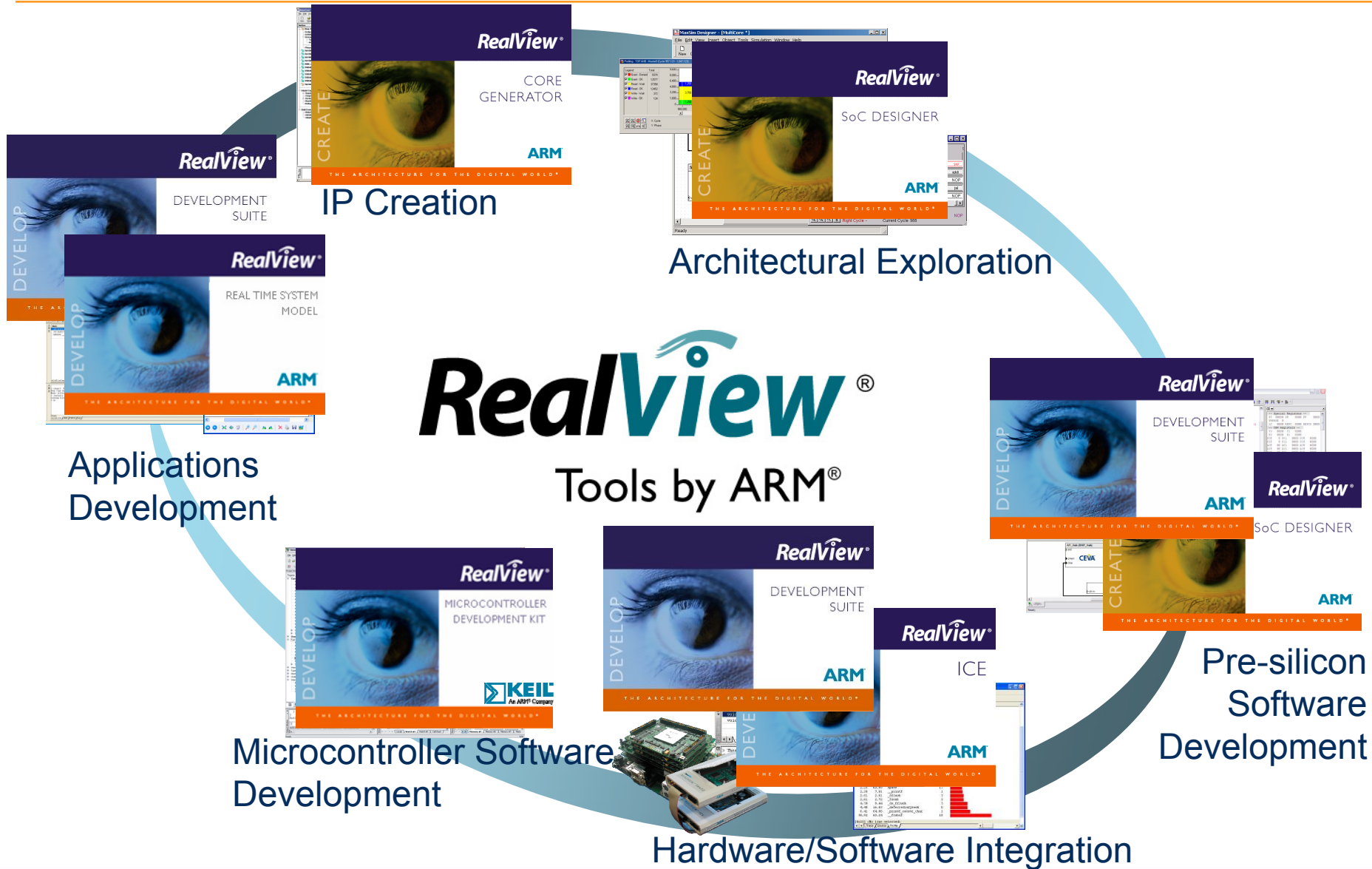


Pre-silicon Software Development



Hardware/Software Integration

Comprehensive Tools

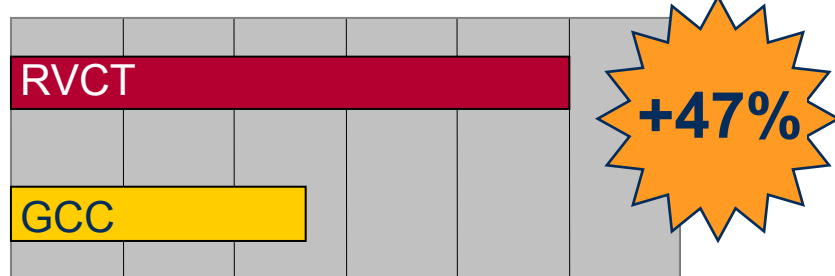


RealView® Tools for Linux Targets

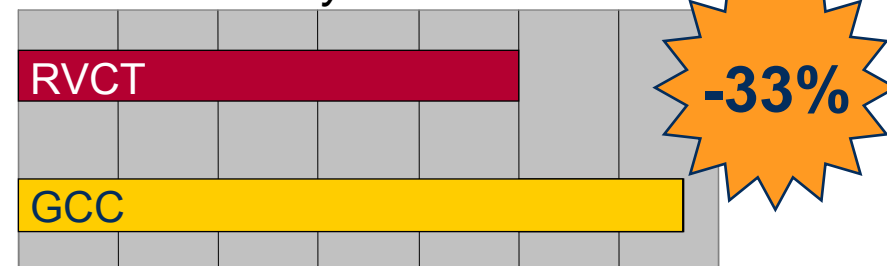
- Easy to use, simplified debug and visualization of Linux kernel and applications on multiple CPUs
- Use GDB and GCC interchangeably with RealView Tools for Linux development
- Save memory and improve performance by using RealView tools



Code Performance: RVCT vs. GCC



Code Density: RVCT vs. GCC

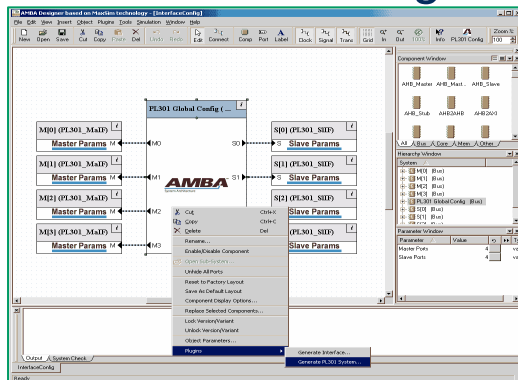


Right-First-Time Design

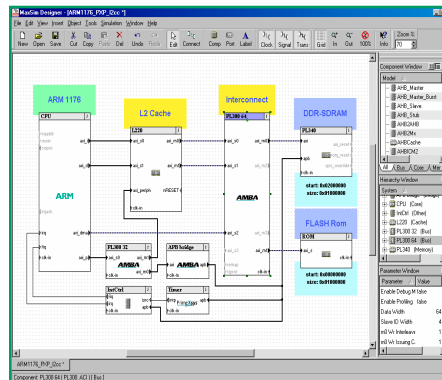
- AMBA[®] Designer
 - Automating the design of AMBA 3 AXI[™]-based interconnect, fabric and sub-systems
- Direct RTL implementation
- The means to move from specification to implementation
- Common entry point for all PrimeCell[®] IP



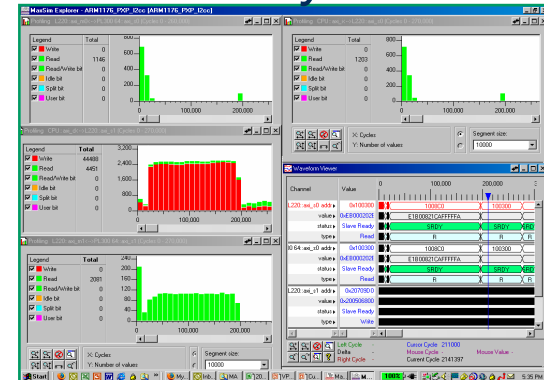
Construct & Configure



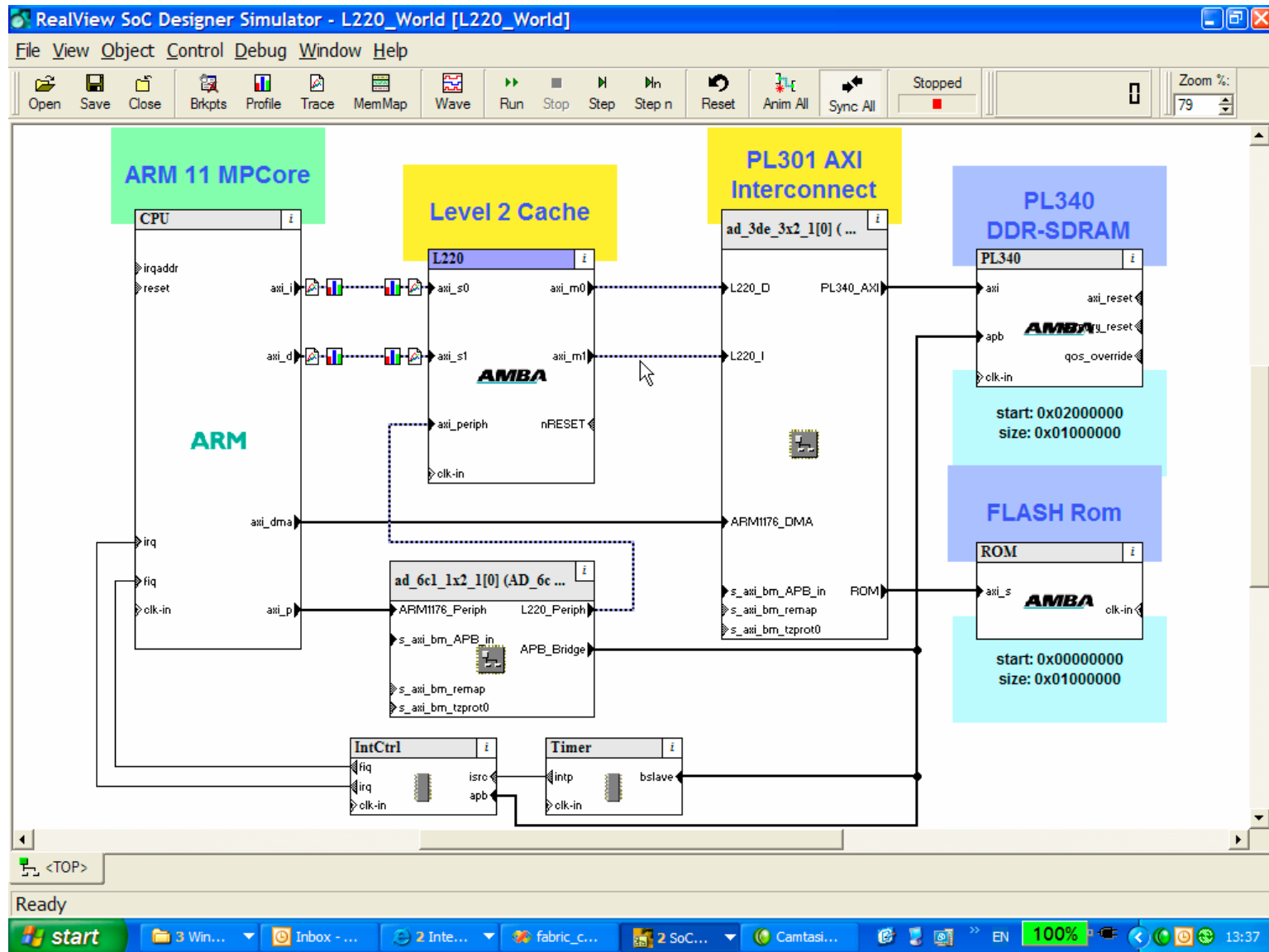
Simulate



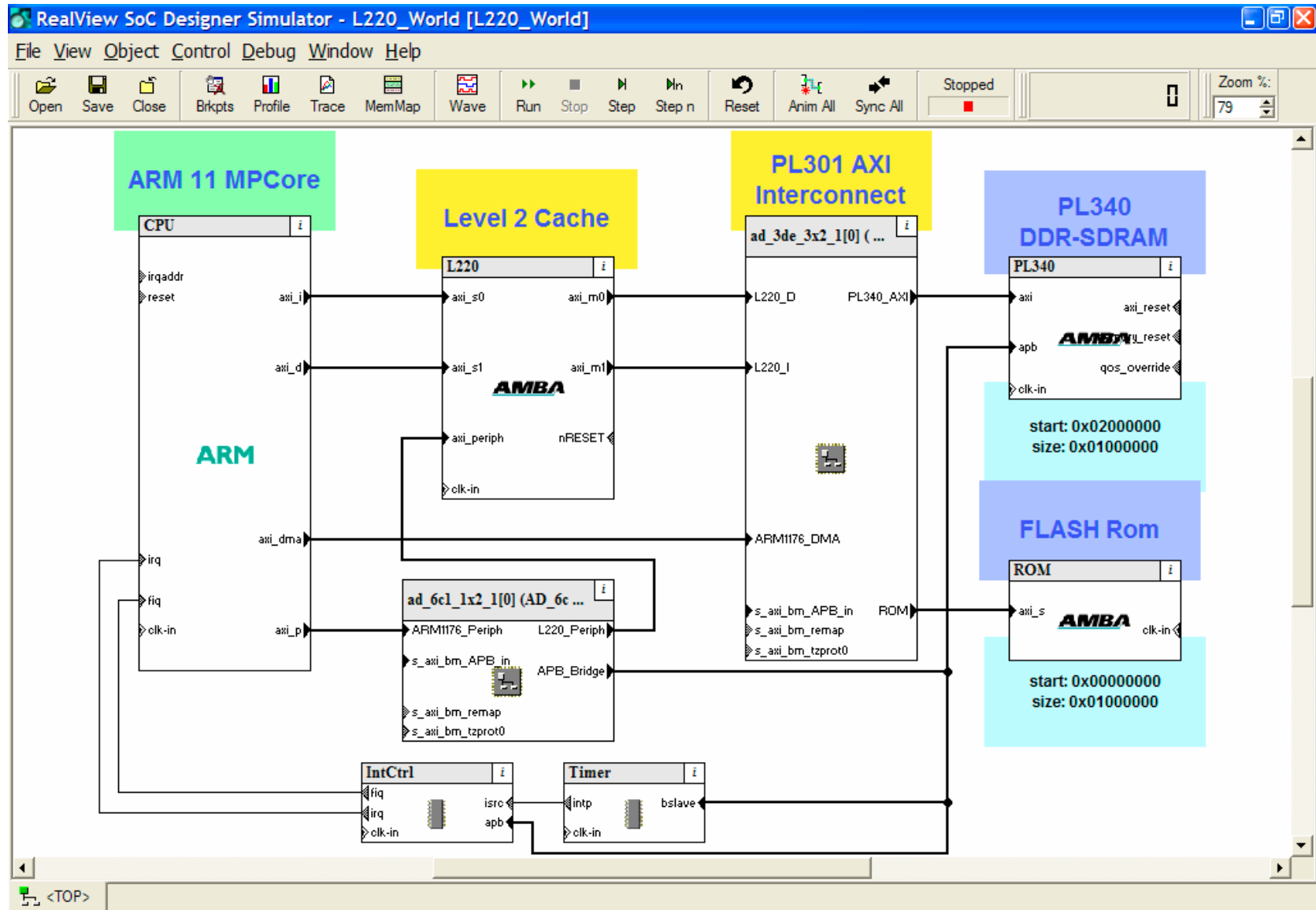
Analyse



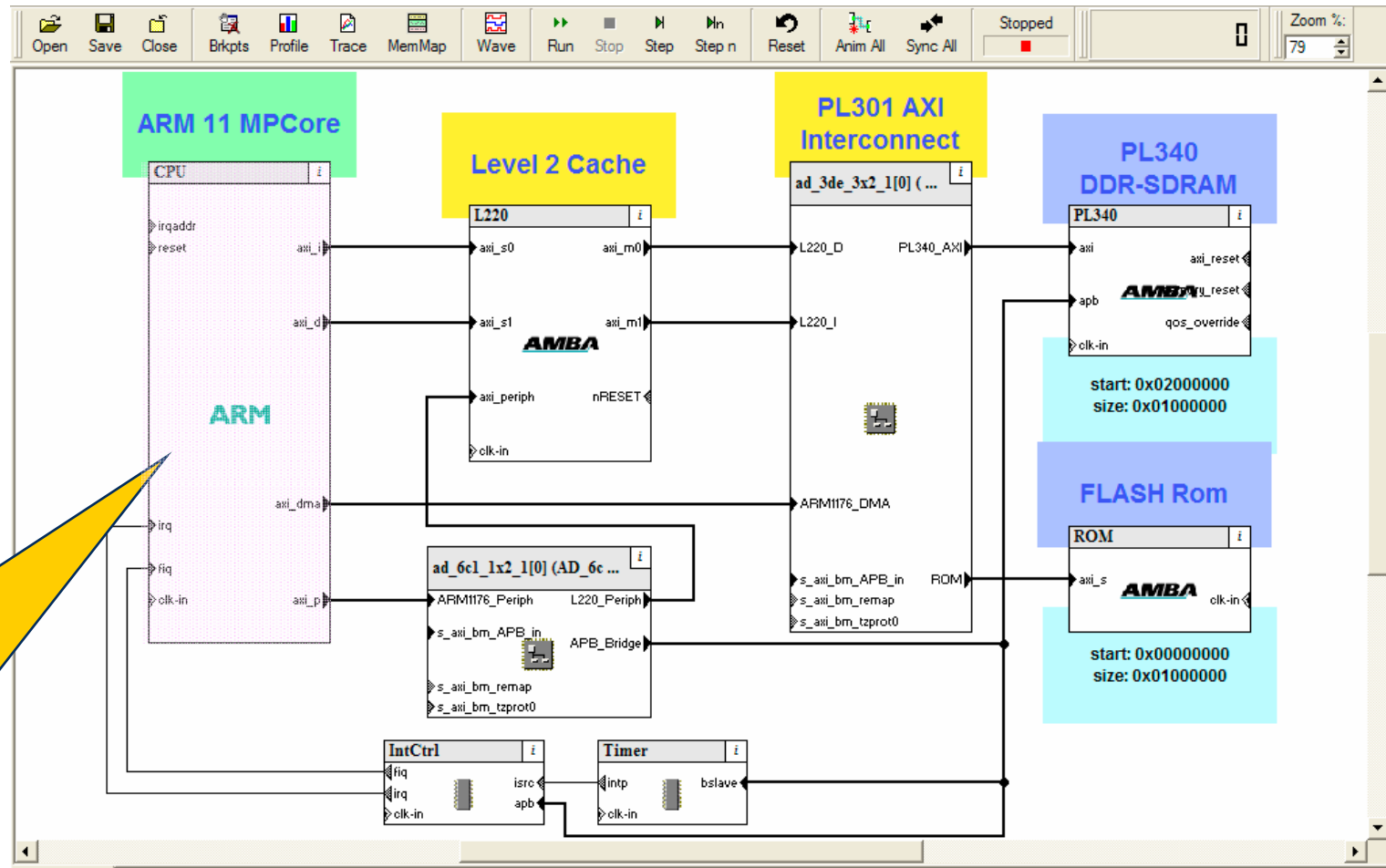
AMBA Designer



AMBA Designer



Scalable Performance



CPU with throughput and scalability to meet multiple performance points and give design headroom

Bandwidth and Security Drive CPUs

- Content consumption by the consumer is driving bandwidth requirements at a faster rate than ever before
 - Typically need 100Mbps to support multiple concurrent HDTV streams to the home plus data, plus voice....
 - ⇒ More CPU performance required
- Gateways involve more Network Layers than before
 - VPN, Firewall, Anti-Virus, Encryption, NAT require more “Payload Manipulation”
 - No longer simply moving data from A-to-B based on packet header
 - Intelligent packet processing is needed
 - ⇒ More CPU performance required
- The Challenge: Change the economics of system design
 - Deliver the performance of Carrier Class systems, at a cost suitable for consumer-level deployment

ARM11™ MPCore™

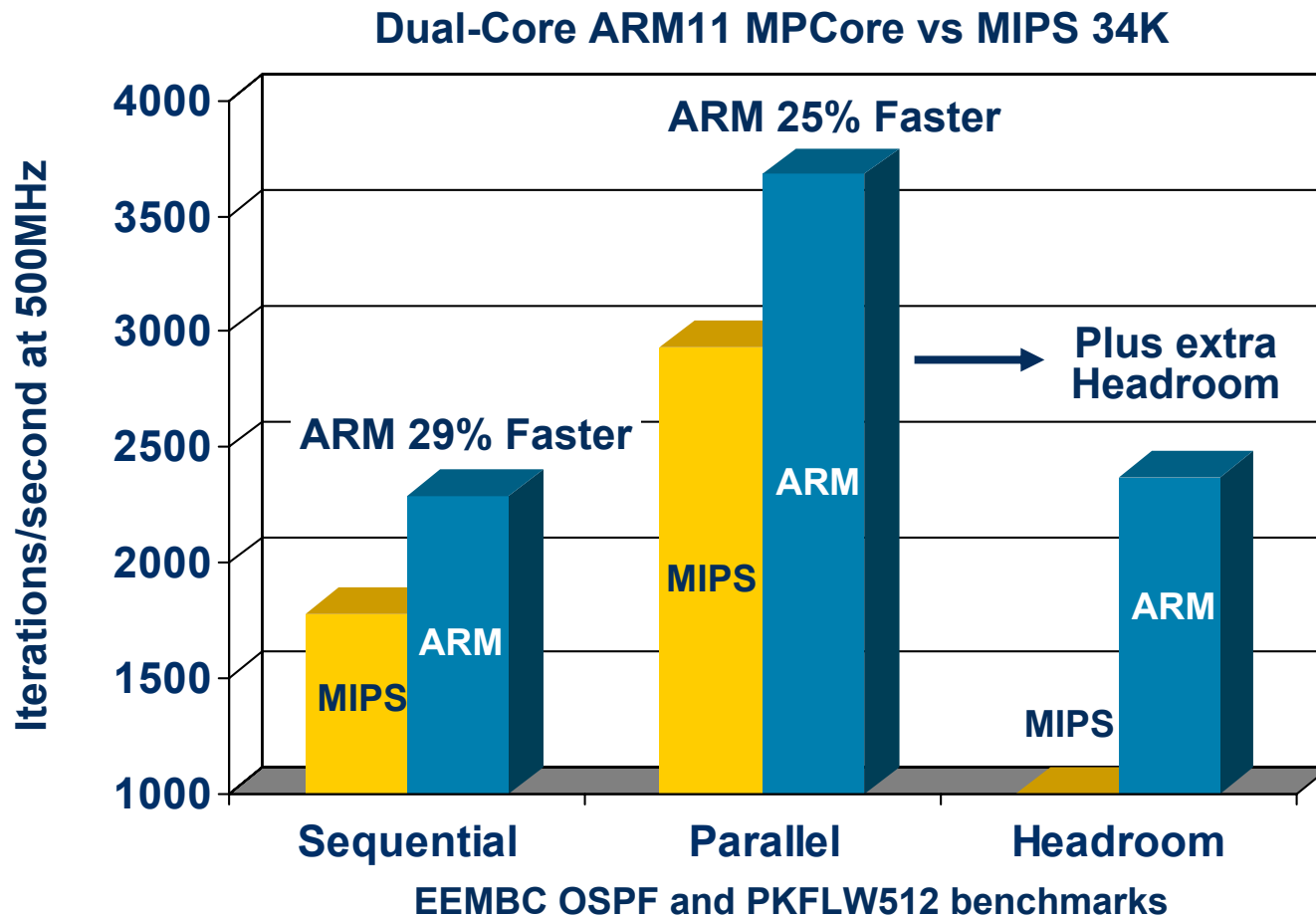
- Scalable processor for high performance AND low power
 - Silicon since May 2005
- Optimised memory system for throughput and cache coherence for full SMP support



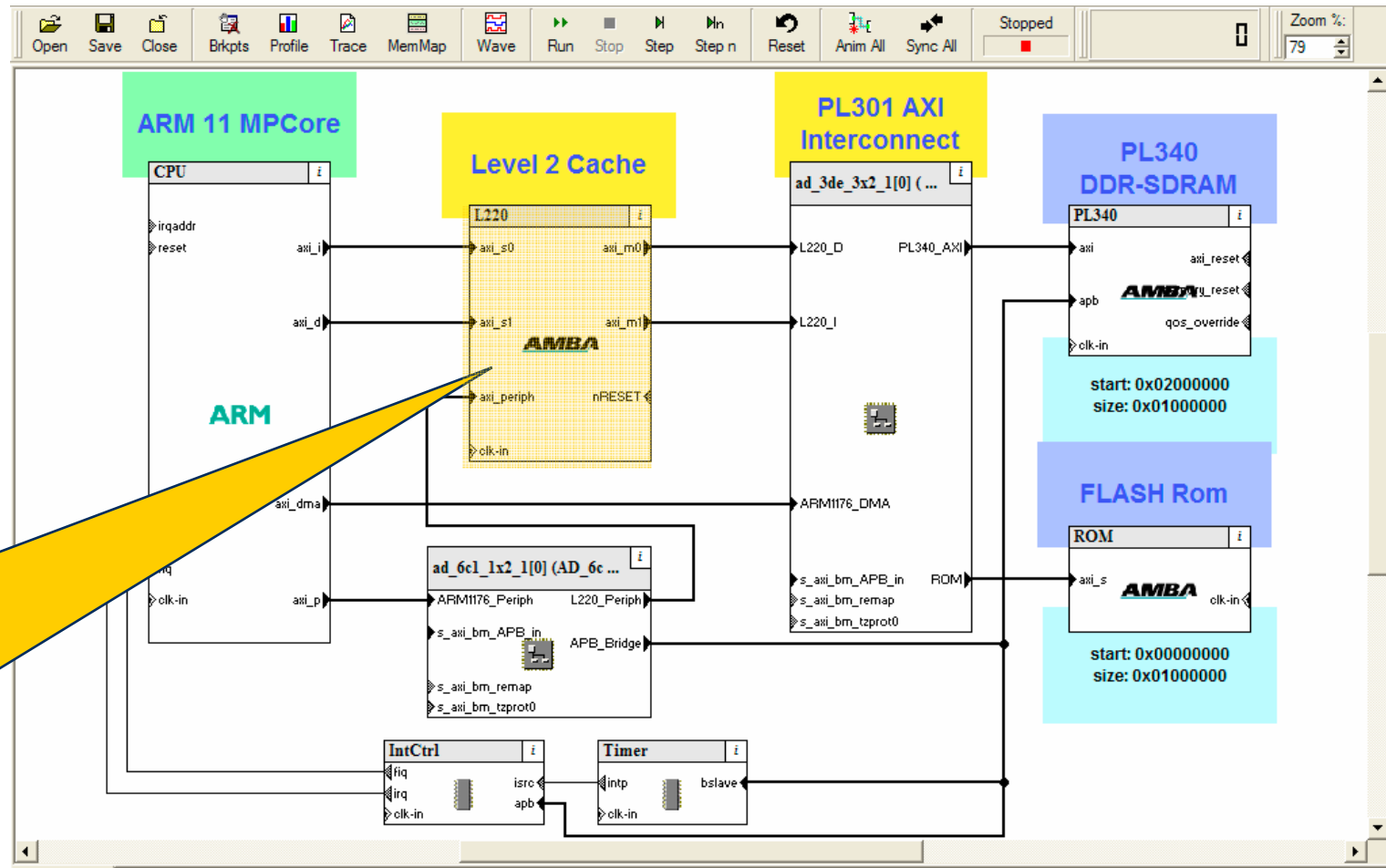
TSMC 90G process	MIPS 34K	ARM MPCore (x2)	ARM Advantage
Speed (MHz)	500	800 (2x 400) (targeted speed)	60% faster
Area ² (mm ²) (inc. 32K cache)	4.7	3.6	31% smaller
Power ² (mW) excluding rams	295	230	28% lower
DSP Benchmarks (est)	~1,490	2,812	88% faster
Compatibility	Need MT modifications	Traditional SMP or dual (AMP) OS	Time to market

Higher Performance, Better Throughput

- Additional threads run on a the second CPU at full speed
 - MPCore gives additional headroom for even greater speedups



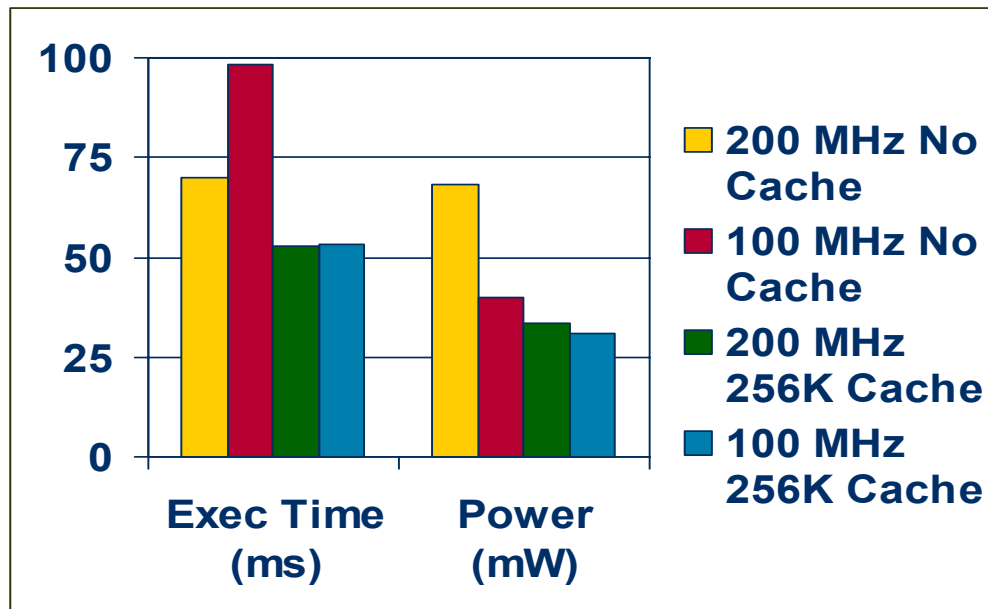
Fabric IP: Extending Core Performance



Efficient memory system to ensure sufficient bandwidth available to main "fast path" dataflow

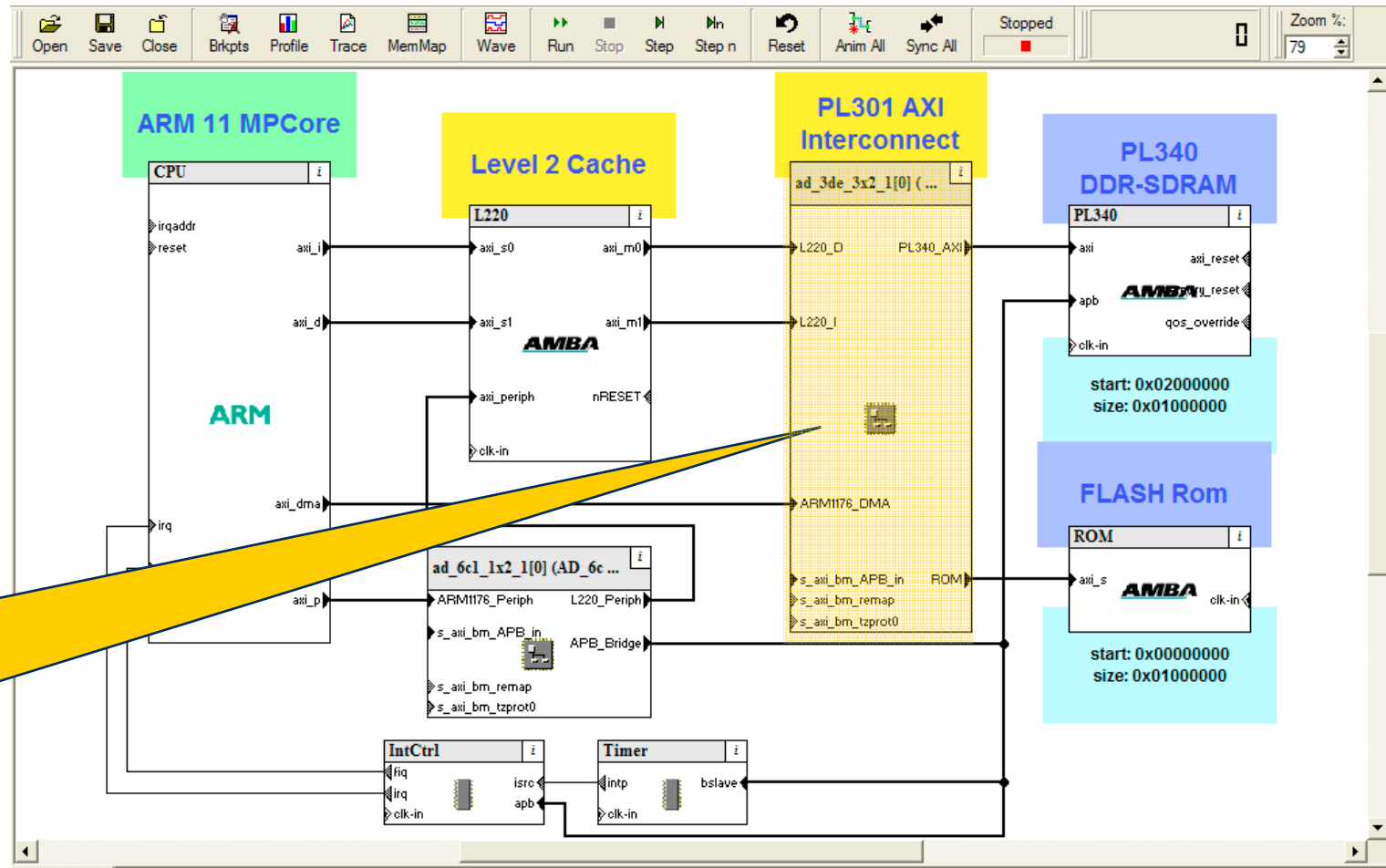
Extending Core Performance: L2 Cache

- AMBA 3 AXI and TrustZone[®] support:
 - Easy integration into ARM-based system
- Configurable size - Up to 2M bytes
 - Optimise Performance/Area trade-off
- Uses standard compiled RAMs
 - Widest possible choice of RAM technology



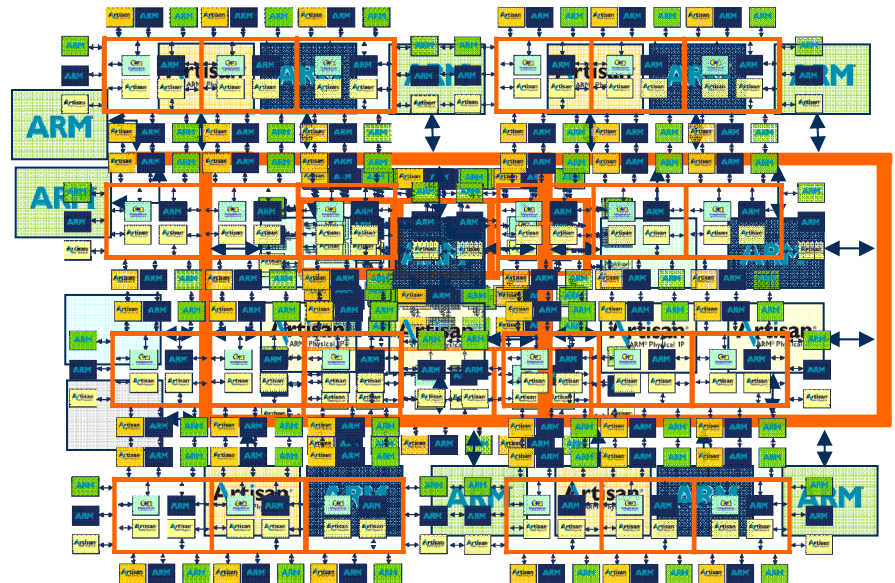
Addition of L2 Cache
increases performance
and
reduces system power

Fabric IP: End-to-end QoS

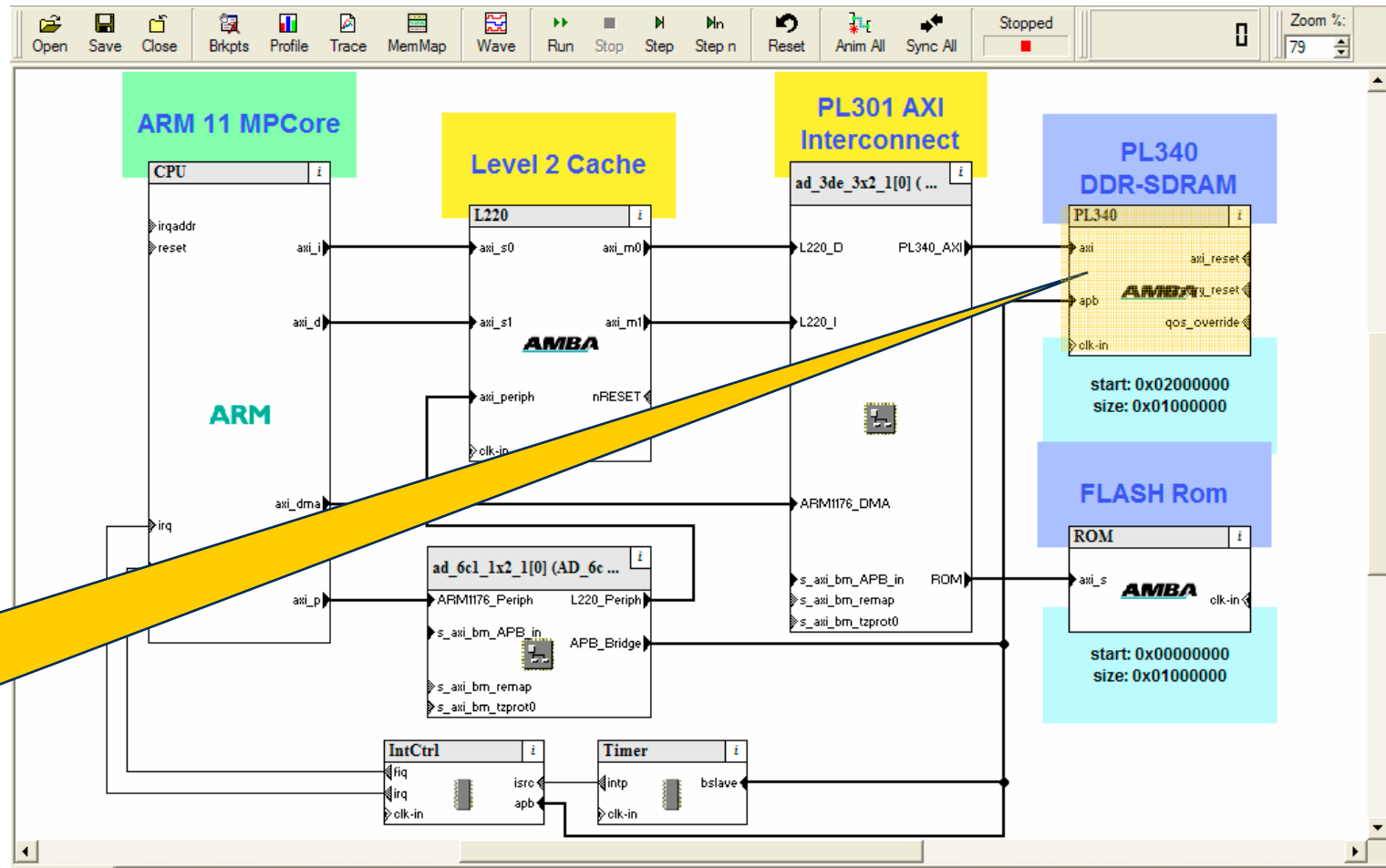


Extending SoC Performance

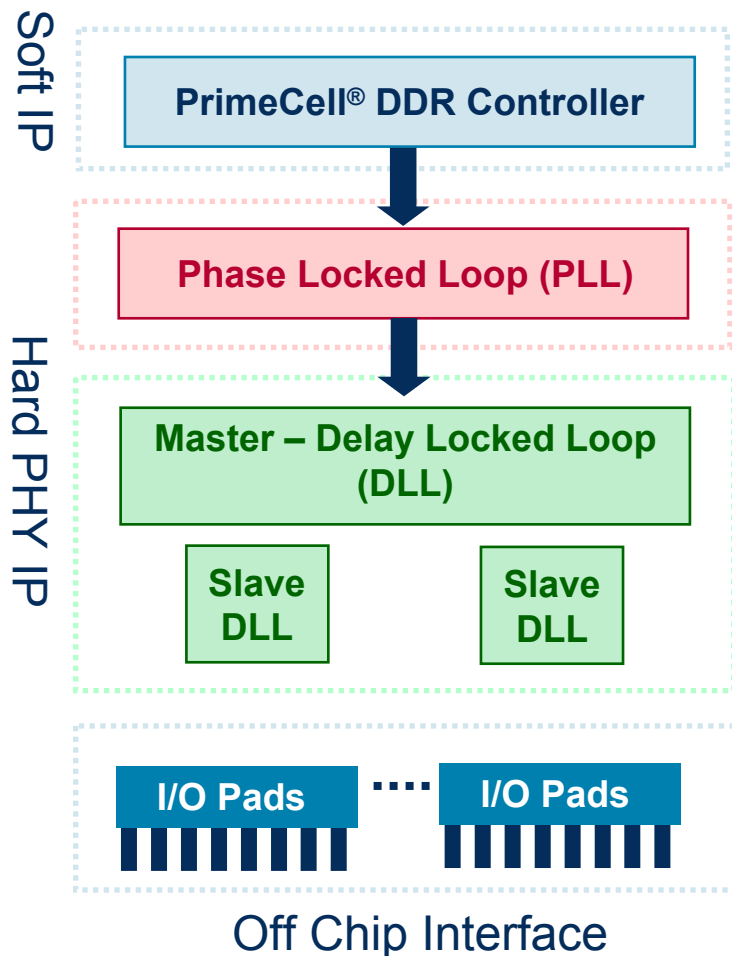
- AMBA - the *de facto* standard
- Complete on-chip communications fabric
- Optimised performance from CPU to Memory
 - No bottlenecks
- Highly configurable
 - e.g. QoS scheme
 - AMBA 3 AXI/AHB™/APB™
- Ease of integration
- Low risk solution



Fabric IP, DRAM interface

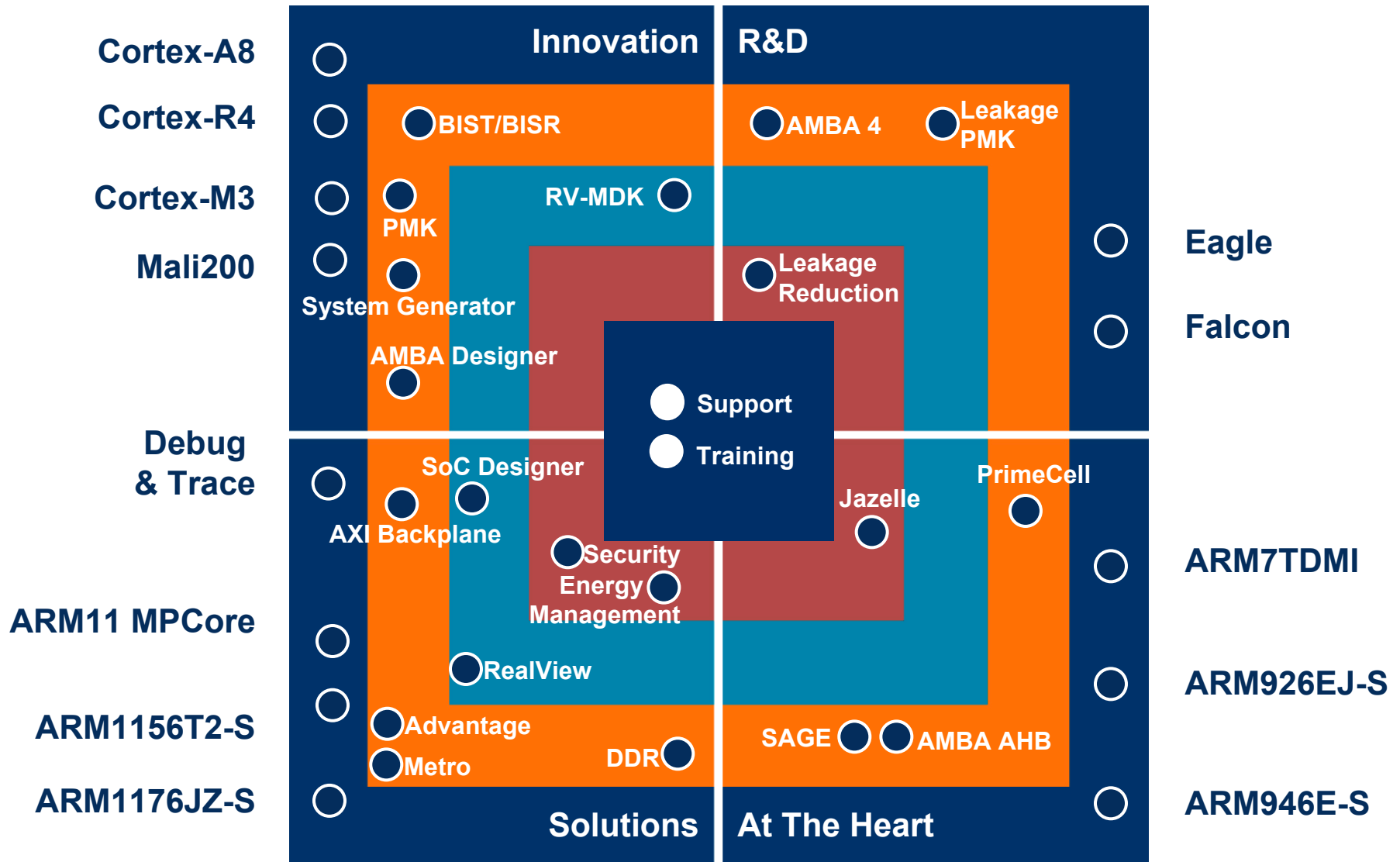


Flexible DDR Solution



- ARM PrimeCell® DDR controllers
 - AXI DDR/MDDR (PL340) -Now
 - AHB DDR (PL24x) - Now
 - AXI DDR2 (PL341) - Available Oct '06
- Flexible DDR Interface optimisation
 - Area, Power and Performance
- DDR Interfaces support
 - DDR, DDR2, MDDR GDDR3 - Now
 - DDR3 - Q406
- DDR Interface features
 - Dynamic calibrator for on-die termination
 - Pull-up/down network for PVT variations

Enterprise



Critical Success Factors

- Cost Effective
 - Power efficiency needed for package cost and fan elimination
- Standard S/W Infrastructure
 - Complete Networking Stack, including NAT, IPSec, Firewall,
- Compiler and Debug Tools
 - Development tools strongly integrated with Operating System
- Scalable Performance
 - Design headroom for evolving standards
- End-to-end QoS
 - Intelligent fabric to ensure traffic prioritisation
- Right-first-time Design
 - Extensive pre-silicon design validation of “random” packet flows and conflicts through system modeling

